

SARDAR PATEL UNIVERSITY
B.Sc (CA&IT) Integrated / M.Sc IT Integrated (NC) Examination
3rd Semester (CBCS)
Tuesday , Date: 23rd October, 2018
Session : Morning Time : 10:00 AM to 01:00 PM

Course Code: PS03CHIT01

Course Title : Operating Systems - I

Total Marks: 70

Q1. Multiple Choice Questions.

[10]

1. _____ is a program in execution.

a. Job	c. Process
b. Kernel	d. File
2. _____ is the hidden part of operating system.

a. Job	c. Process
b. Kernel	d. File
3. Which of the following is not a state of a process?

a. Ready	c. Interrupt
b. Running	d. New
4. _____ indicates the address of the next instruction to be executed.

a. Program state .	c. Program counter
b. Process state	d. Process counter
5. _____ is the sum of the periods spent waiting in the ready queue.

a. Turnaround time	c. Throughput
b. Waiting time	d. Response time
6. _____ is the solution of the bounded buffer.

a. Shared-Memory	c. Data consistency
b. Virtual Memory	d. None
7. Directed graph, in which deadlock can be described called _____

a. Resources Allocation graph	c. Non-preemption Graph
b. Allocation graph	d. Request Allocation Graph
8. _____ refers to an interface design for connecting peripheral devices.

a. Transfer rate	c. Firewire
b. Latency time	d. Disk controller
9. The simplest approach to introducing redundancy is to duplicate every disk is called _____.

a. Mirroring	c. Striping
b. Parity	d. None of these
10. What is the full form of HBA?

a. Hub Bus Adapter	c. Host Bus Adapter
b. Host byte Adapter	d. Hub Bit Adapter

Q2. Answer the following short questions (Attempt any TEN)

[20]

1. What is operating system?
2. Briefly explain the terms: (1) Shell and (2) Trap
3. Briefly explain user mode and kernel mode of operating system.

4. Draw process state diagram.
5. Briefly explain difference between device queue and ready queue.
6. What is base register and limit register?
7. What does flag and turn indicated in Peterson's solution?
8. What is spinlock?
9. Explain race condition for concurrent processes.
10. State advantages of RAID level 3 over level 1.
11. What are controllers? List its types.
12. What is disk bandwidth? How it can be improved?

- Q3.a. Write a note on virtual machines. [5]
 b. Explain the concept of real time operating systems. [5]

OR

- Q3.a. Explain various operations of Operating System. [5]
 b. Explain Client Server model in distributed system in detail. [5]

- Q4.a. What is priority scheduling? What is the drawback of priority scheduling? [5]
 What is the solution for the drawback?
 b. Explain in detail various scheduling criteria. [5]

OR

- Q4.a. Explain concept of process scheduling with help of Queuing diagram. [5]
 b. What is PCB? Draw and explain the structure of PCB. [5]
 Q5.a. What is semaphore? Explain types of semaphores with their usage [5]
 b. Write note on deadlock prevention. [5]

OR

- Q5.a. What is deadlock? Explain necessary conditions for occurrence of deadlock [5]
 b. What is critical section problem? Explain requirements for solution to critical section problem. [5]

- Q6. What is RAID? Explain RAID levels in detail [10]

OR

- Q6. Explain the idea behind SSTF disk scheduling algorithm. Consider following [10]
 queue of requests and calculate total movement of head in terms of number of
 cylinders by using SSTF algorithm.
 Queue = 98, 183, 37, 122, 14, 124, 65, 67
 Head starts at 53

— X —

(2)